

Database Management System

CSIT — 4th Semester — 2080 — Answer Sheet
Subject Code: CSC265

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Consider a banking database with three tables and primary keys underlined as given below: Customer(CustomerID, CustomerName, Address, Phone, Email) Owns(CustomerID, AccountNumber) Account(AccountNumber, AccountType, Balance) Write both relational algebra and SQL queries: a. To display name of all customers who live in "Kathmandu". b. To count total number of customers. c. To find name of those customers who have balance greater than or equal to 100000. d. To find average balance of each account type.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define normalization. Why normalization is important in database design? Explain 1NF, 2NF, and 3NF with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is two-phase locking? What are different types of locks in two-phase locking? Explain basic, conservative, strict, and rigorous two-phase locking. What is lock conversion?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. What is flat-file system? What are advantages of using DBMS approach?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define data abstraction, data model, schemas, instances, and database state.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is conceptual data model? Explain different types of attributes used in ER diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is relational model? Define the terms domain, attribute, tuple, and relation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is tuple relational calculus? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define transaction. What are different desirable properties of transaction?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Why do we need concurrency control in databases? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Why database recovery is essential? Explain recovery technique based on immediate update.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: a. Natural join b. Shadow paging

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.